PIXEL CHAMP

THE ULTIMATE DATA SCIENCE, MACHINE LEARNING & GENAI INTERVIEW QUESTION BANK

Train Smart. Prioritize What Matters. Win the Interview.

1,443	10	70	100%
UNIQUE QUESTIONS	MAJOR PARTS	TECHNICAL DOMAINS	CURATED & DEDUPED

PRIORITY TIERS: ★★★ HIGH (Core) | ★★■ MEDIUM (Applied) | ★■■ LOWER (Niche)

[OFFICIAL GITHUB REPOSITORY -- FULL ANSWERS & CODE]
https://github.com/bsaclubmnnit-bot/ML-DS-QUESTIONS

Concept, Research & Architecture | 2026 Production Master Handbook | Interactive PDF Navigation

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Complete 10-Part Taxonomy & Master Topic Index (Click any topic row to jump directly to it)

GitHub Answers & Code: https://github.com/bsaclubmnnit-bot/ML-DS-QUESTIONS

DOMAIN / TOPIC TITLE	QUESTIONS	PRIORITY SUMMARY	JUMP LINK
PART I: DATA SCIENCE FOUNDATIONS			
Topic 1: Statistics and Descriptive Statistics	**71**	60 ★★★ | 8 ★★■ | 3 ★■■	GO TO TOPIC >>
Topic 2: Probability and Probability Distributions	**17**	17 ★★★ | 0 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 3: Hypothesis Testing and Statistical Inference	**20**	15 ★★★ | 5 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 4: Exploratory Data Analysis (EDA)	**44**	10 ★★★ | 33 ★★■ | 1 ★■■	GO TO TOPIC >>
Topic 5: Feature Engineering and Feature Selection	**49**	37 ★★★ | 10 ★★■ | 2 ★■■	GO TO TOPIC >>
Topic 6: Data Preprocessing and Data Leakage	**11**	6 ★★★ | 5 ★★■ | 0 ★■■	GO TO TOPIC >>
PART II: SQL, PYTHON AND DATA ANALYSIS			
Topic 7: SQL Fundamentals and Database Concepts	**115**	77 ★★★ | 38 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 8: SQL Joins, Subqueries and Set Operations	**19**	15 ★★★ | 4 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 9: SQL Aggregations, CTEs and Window Functions	**17**	12 ★★★ | 5 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 10: Advanced SQL and Business Analytics Problems	**7**	0 ★★★ | 7 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 11: Python Fundamentals and Data Structures	**58**	52 ★★★ | 6 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 12: Python OOP, Iterators, Generators and Decorators	**12**	10 ★★★ | 2 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 13: Python Performance, Concurrency and Memory	**8**	6 ★★★ | 1 ★★■ | 1 ★■■	GO TO TOPIC >>
Topic 14: NumPy and Vectorized Computation	**5**	4 ★★★ | 1 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 15: Pandas and Data Manipulation	**57**	33 ★★★ | 24 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 16: Data Cleaning and Analytical Pipelines	**3**	0 ★★★ | 3 ★★■ | 0 ★■■	GO TO TOPIC >>
PART III: MACHINE LEARNING			
Topic 17: Supervised and Unsupervised Learning	**31**	20 ★★★ | 9 ★★■ | 2 ★■■	GO TO TOPIC >>
Topic 18: Linear and Logistic Regression	**35**	24 ★★★ | 11 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 19: Regularization: Ridge, Lasso and Elastic Net	**24**	10 ★★★ | 14 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 20: Naive Bayes and K-Nearest Neighbors	**25**	12 ★★★ | 11 ★★■ | 2 ★■■	GO TO TOPIC >>
Topic 21: Decision Trees	**22**	11 ★★★ | 10 ★★■ | 1 ★■■	GO TO TOPIC >>
Topic 22: Ensemble Learning, Bagging and Boosting	**14**	10 ★★★ | 4 ★★■ | 0 ★■■	GO TO TOPIC >>

Topic 23: Random Forest	**13**	4 ★★★ | 8 ★★■ | 1 ★■■	GO TO TOPIC >>
Topic 24: Gradient Boosting, AdaBoost and XGBoost	**34**	15 ★★★ | 19 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 25: Clustering: K-Means, Hierarchical Clustering and DBSCAN	**45**	19 ★★★ | 23 ★★■ | 3 ★■■	GO TO TOPIC >>
Topic 26: Bias-Variance Tradeoff and Generalization	**12**	6 ★★★ | 6 ★★■ | 0 ★■■	GO TO TOPIC >>
PART IV: MODEL EVALUATION AND PRODUCTION ML			
Topic 27: Classification Metrics and Confusion Matrix	**33**	28 ★★★ | 5 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 28: Regression Metrics	**2**	1 ★★★ | 0 ★★■ | 1 ★■■	GO TO TOPIC >>
Topic 29: ROC-AUC, PR-AUC and Threshold Optimization	**8**	5 ★★★ | 3 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 30: Imbalanced Classification	**2**	1 ★★★ | 1 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 31: Cross-Validation and Hyperparameter Tuning	**11**	10 ★★★ | 1 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 32: Calibration, Robustness and Model Selection	**4**	3 ★★★ | 0 ★★■ | 1 ★■■	GO TO TOPIC >>
Topic 33: ML Pipelines and Data Leakage Prevention	**2**	2 ★★★ | 0 ★★■ | 0 ★■■	GO TO TOPIC >>
PART V: DEEP LEARNING			
Topic 34: Neural Networks and Backpropagation	**45**	39 ★★★ | 5 ★★■ | 1 ★■■	GO TO TOPIC >>
Topic 35: RNNs, LSTM, GRU and BiLSTM	**32**	10 ★★★ | 16 ★★■ | 6 ★■■	GO TO TOPIC >>
Topic 36: Attention and Transformer Architecture	**22**	16 ★★★ | 6 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 37: Embeddings and Representation Learning	**1**	1 ★★★ | 0 ★★■ | 0 ★■■	GO TO TOPIC >>
PART VI: GENERATIVE AI AND LLMs			
Topic 38: Generative AI and LLM Fundamentals	**29**	23 ★★★ | 6 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 39: Tokenization, Embeddings and Context Windows	**7**	6 ★★★ | 1 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 40: Prompt Engineering and Context Engineering	**26**	15 ★★★ | 11 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 41: LLM Inference, Decoding and Sampling	**4**	3 ★★★ | 1 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 42: Fine-Tuning, PEFT, LoRA and QLoRA	**34**	18 ★★★ | 16 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 43: RLHF, DPO and Model Alignment	**12**	9 ★★★ | 3 ★★■ | 0 ★■■	GO TO TOPIC >>
PART VII: RAG AND INFORMATION RETRIEVAL			
Topic 44: RAG Fundamentals and Architecture	**19**	15 ★★★ | 4 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 45: Document Parsing, Chunking and Indexing	**9**	5 ★★★ | 4 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 46: Embedding Models and Vector Databases	**28**	19 ★★★ | 8 ★★■ | 1 ★■■	GO TO TOPIC >>

Topic 47: Sparse Retrieval, BM25 and Hybrid Search	**2**	1 ★★★ | 1 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 48: Reranking and Advanced Retrieval	**2**	1 ★★★ | 1 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 49: Advanced RAG, GraphRAG and Multi-Hop RAG	**31**	16 ★★★ | 15 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 50: RAG Evaluation and Troubleshooting	**31**	15 ★★★ | 15 ★★■ | 1 ★■■	GO TO TOPIC >>
Topic 51: Hallucination Reduction, Grounding and Citations	**2**	2 ★★★ | 0 ★★■ | 0 ★■■	GO TO TOPIC >>
PART VIII: AGENTIC AI AND ORCHESTRATION			
Topic 52: AI Agent Fundamentals	**30**	21 ★★★ | 9 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 53: Tool Calling and Structured Outputs	**19**	3 ★★★ | 16 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 54: MCP and Tool Integration	**8**	4 ★★★ | 4 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 55: LangChain Framework	**20**	16 ★★★ | 4 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 56: LangGraph, State, Nodes, Edges and Checkpointing	**6**	3 ★★★ | 3 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 57: Multi-Agent Systems and Orchestration	**16**	7 ★★★ | 8 ★★■ | 1 ★■■	GO TO TOPIC >>
Topic 58: Agent Memory and Context Management	**14**	3 ★★★ | 11 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 59: AI Safety, Security and Guardrails	**24**	9 ★★★ | 15 ★★■ | 0 ★■■	GO TO TOPIC >>
PART IX: LLMOPS AND AI SYSTEM DESIGN			
Topic 60: LLMOps and Observability	**20**	9 ★★★ | 10 ★★■ | 1 ★■■	GO TO TOPIC >>
Topic 61: Evaluation Frameworks and Testing	**1**	0 ★★★ | 1 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 62: Latency, Token Costs, Caching and Optimization	**3**	0 ★★★ | 3 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 63: Reliability, Retries and Production Monitoring	**4**	1 ★★★ | 3 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 64: Model Routing and Inference Architecture	**3**	2 ★★★ | 1 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 65: End-to-End RAG System Design	**8**	0 ★★★ | 8 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 66: Agentic Workflow System Design	**10**	0 ★★★ | 10 ★★■ | 0 ★■■	GO TO TOPIC >>
Topic 67: Enterprise AI Architecture and Scalability	**9**	0 ★★★ | 9 ★★■ | 0 ★■■	GO TO TOPIC >>
PART X: PRACTICAL INTERVIEW ROUNDS			
Topic 68: Coding and Implementation Questions	**20**	1 ★★★ | 18 ★★■ | 1 ★■■	GO TO TOPIC >>
Topic 69: Scenario-Based Data Science Questions	**25**	2 ★★★ | 11 ★★■ | 12 ★■■	GO TO TOPIC >>
Topic 70: Rapid-Fire Comparisons	**37**	37 ★★★ | 0 ★★■ | 0 ★■■	GO TO TOPIC >>

PRIORITY QUICK-REFERENCE INDEX

High-Yield Interview Roadmap: Core Foundational Questions (★★★) Grouped by Technical Domain

TECHNICAL DOMAIN	TOTAL QUESTIONS	★★★ HIGH PRIORITY	DIRECT ACCESS
Topic 1: Statistics and Descriptive Statistics	71	**60 Questions**	JUMP TO TOPIC >>
Topic 2: Probability and Probability Distributions	17	**17 Questions**	JUMP TO TOPIC >>
Topic 3: Hypothesis Testing and Statistical Inference	20	**15 Questions**	JUMP TO TOPIC >>
Topic 4: Exploratory Data Analysis (EDA)	44	**10 Questions**	JUMP TO TOPIC >>
Topic 5: Feature Engineering and Feature Selection	49	**37 Questions**	JUMP TO TOPIC >>
Topic 6: Data Preprocessing and Data Leakage	11	**6 Questions**	JUMP TO TOPIC >>
Topic 7: SQL Fundamentals and Database Concepts	115	**77 Questions**	JUMP TO TOPIC >>
Topic 8: SQL Joins, Subqueries and Set Operations	19	**15 Questions**	JUMP TO TOPIC >>
Topic 9: SQL Aggregations, CTEs and Window Functions	17	**12 Questions**	JUMP TO TOPIC >>
Topic 10: Advanced SQL and Business Analytics Problems	7	**0 Questions**	JUMP TO TOPIC >>
Topic 11: Python Fundamentals and Data Structures	58	**52 Questions**	JUMP TO TOPIC >>
Topic 12: Python OOP, Iterators, Generators and Decorators	12	**10 Questions**	JUMP TO TOPIC >>
Topic 13: Python Performance, Concurrency and Memory	8	**6 Questions**	JUMP TO TOPIC >>
Topic 14: NumPy and Vectorized Computation	5	**4 Questions**	JUMP TO TOPIC >>
Topic 15: Pandas and Data Manipulation	57	**33 Questions**	JUMP TO TOPIC >>
Topic 16: Data Cleaning and Analytical Pipelines	3	**0 Questions**	JUMP TO TOPIC >>
Topic 17: Supervised and Unsupervised Learning	31	**20 Questions**	JUMP TO TOPIC >>
Topic 18: Linear and Logistic Regression	35	**24 Questions**	JUMP TO TOPIC >>
Topic 19: Regularization: Ridge, Lasso and Elastic Net	24	**10 Questions**	JUMP TO TOPIC >>
Topic 20: Naive Bayes and K-Nearest Neighbors	25	**12 Questions**	JUMP TO TOPIC >>
Topic 21: Decision Trees	22	**11 Questions**	JUMP TO TOPIC >>
Topic 22: Ensemble Learning, Bagging and Boosting	14	**10 Questions**	JUMP TO TOPIC >>
Topic 23: Random Forest	13	**4 Questions**	JUMP TO TOPIC >>
Topic 24: Gradient Boosting, AdaBoost and XGBoost	34	**15 Questions**	JUMP TO TOPIC >>
Topic 25: Clustering: K-Means, Hierarchical Clustering and DBSCAN	45	**19 Questions**	JUMP TO TOPIC >>
Topic 26: Bias-Variance Tradeoff and Generalization	12	**6 Questions**	JUMP TO TOPIC >>
Topic 27: Classification Metrics and Confusion Matrix	33	**28 Questions**	JUMP TO TOPIC >>
Topic 28: Regression Metrics	2	**1 Questions**	JUMP TO TOPIC >>

Topic 29: ROC-AUC, PR-AUC and Threshold Optimization	8	**5 Questions**	JUMP TO TOPIC >>
Topic 30: Imbalanced Classification	2	**1 Questions**	JUMP TO TOPIC >>
Topic 31: Cross-Validation and Hyperparameter Tuning	11	**10 Questions**	JUMP TO TOPIC >>
Topic 32: Calibration, Robustness and Model Selection	4	**3 Questions**	JUMP TO TOPIC >>
Topic 33: ML Pipelines and Data Leakage Prevention	2	**2 Questions**	JUMP TO TOPIC >>
Topic 34: Neural Networks and Backpropagation	45	**39 Questions**	JUMP TO TOPIC >>
Topic 35: RNNs, LSTM, GRU and BiLSTM	32	**10 Questions**	JUMP TO TOPIC >>
Topic 36: Attention and Transformer Architecture	22	**16 Questions**	JUMP TO TOPIC >>
Topic 37: Embeddings and Representation Learning	1	**1 Questions**	JUMP TO TOPIC >>
Topic 38: Generative AI and LLM Fundamentals	29	**23 Questions**	JUMP TO TOPIC >>
Topic 39: Tokenization, Embeddings and Context Windows	7	**6 Questions**	JUMP TO TOPIC >>
Topic 40: Prompt Engineering and Context Engineering	26	**15 Questions**	JUMP TO TOPIC >>
Topic 41: LLM Inference, Decoding and Sampling	4	**3 Questions**	JUMP TO TOPIC >>
Topic 42: Fine-Tuning, PEFT, LoRA and QLoRA	34	**18 Questions**	JUMP TO TOPIC >>
Topic 43: RLHF, DPO and Model Alignment	12	**9 Questions**	JUMP TO TOPIC >>
Topic 44: RAG Fundamentals and Architecture	19	**15 Questions**	JUMP TO TOPIC >>
Topic 45: Document Parsing, Chunking and Indexing	9	**5 Questions**	JUMP TO TOPIC >>
Topic 46: Embedding Models and Vector Databases	28	**19 Questions**	JUMP TO TOPIC >>
Topic 47: Sparse Retrieval, BM25 and Hybrid Search	2	**1 Questions**	JUMP TO TOPIC >>
Topic 48: Reranking and Advanced Retrieval	2	**1 Questions**	JUMP TO TOPIC >>
Topic 49: Advanced RAG, GraphRAG and Multi-Hop RAG	31	**16 Questions**	JUMP TO TOPIC >>
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Topic 51: Hallucination Reduction, Grounding and Citations	2	**2 Questions**	JUMP TO TOPIC >>
Topic 52: AI Agent Fundamentals	30	**21 Questions**	JUMP TO TOPIC >>
Topic 53: Tool Calling and Structured Outputs	19	**3 Questions**	JUMP TO TOPIC >>
Topic 54: MCP and Tool Integration	8	**4 Questions**	JUMP TO TOPIC >>
Topic 55: LangChain Framework	20	**16 Questions**	JUMP TO TOPIC >>
Topic 56: LangGraph, State, Nodes, Edges and Checkpointing	6	**3 Questions**	JUMP TO TOPIC >>
Topic 57: Multi-Agent Systems and Orchestration	16	**7 Questions**	JUMP TO TOPIC >>
Topic 58: Agent Memory and Context Management	14	**3 Questions**	JUMP TO TOPIC >>

Topic 59: AI Safety, Security and Guardrails	24	**9 Questions**	**JUMP TO TOPIC >>**
Topic 60: LLMOps and Observability	20	**9 Questions**	**JUMP TO TOPIC >>**
Topic 61: Evaluation Frameworks and Testing	1	**0 Questions**	**JUMP TO TOPIC >>**
Topic 62: Latency, Token Costs, Caching and Optimization	3	**0 Questions**	**JUMP TO TOPIC >>**
Topic 63: Reliability, Retries and Production Monitoring	4	**1 Questions**	**JUMP TO TOPIC >>**
Topic 64: Model Routing and Inference Architecture	3	**2 Questions**	**JUMP TO TOPIC >>**
Topic 65: End-to-End RAG System Design	8	**0 Questions**	**JUMP TO TOPIC >>**
Topic 66: Agentic Workflow System Design	10	**0 Questions**	**JUMP TO TOPIC >>**
Topic 67: Enterprise AI Architecture and Scalability	9	**0 Questions**	**JUMP TO TOPIC >>**
Topic 68: Coding and Implementation Questions	20	**1 Questions**	**JUMP TO TOPIC >>**
Topic 69: Scenario-Based Data Science Questions	25	**2 Questions**	**JUMP TO TOPIC >>**
Topic 70: Rapid-Fire Comparisons	37	**37 Questions**	**JUMP TO TOPIC >>**

PART I
DATA SCIENCE FOUNDATIONS

TOPIC 1: STATISTICS AND DESCRIPTIVE STATISTICS

Total: **71 Questions** | High Priority: **60** | Medium: **8** | Lower: **3**

[STAT-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are independent events?

[STAT-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are mutually exclusive events?

[STAT-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are population and sample?

[STAT-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are simple random sampling and stratified sampling?

[STAT-005] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Bonferroni correction?

[STAT-006] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Pearson correlation?

[STAT-007] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Simpson's paradox?

[STAT-008] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Spearman correlation?

[STAT-009] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Z-score?

[STAT-010] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a confidence interval?

[STAT-011] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a confidence level?

[STAT-012] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a confounding variable?

[STAT-013] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a continuous random variable?

[STAT-014] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a discrete random variable?

[STAT-015] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a non-parametric test?

[STAT-016] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a parameter?

[STAT-017] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a percentile?

[STAT-018] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a quantile?

[STAT-019] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a random variable?

[STAT-020] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a statistic?

[STAT-021] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is bootstrapping?

[STAT-022] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is cluster sampling?

[STAT-023] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is coefficient of variation?

[STAT-024] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is confounding?

[STAT-025] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is correlation coefficient range?

[STAT-026] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is correlation matrix?

[STAT-027] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is correlation?

[STAT-028] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is covariance matrix?

[STAT-029] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is covariance?

[STAT-030] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is expected value?

[STAT-031] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is false discovery rate?

[STAT-032] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is kurtosis?

[STAT-033] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is mean?

[STAT-034] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is median?

[STAT-035] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is mode?

[STAT-036] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is negative skewness?

[STAT-037] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is permutation testing?

[STAT-038] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is positive skewness?

[STAT-039] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is range?

[STAT-040] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is sampling bias?

[STAT-041] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is sampling?

[STAT-042] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is selection bias?

[STAT-043] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is skewness?

[STAT-044] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is standard deviation?

[STAT-045] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is statistical power?

[STAT-046] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is survivorship bias?

[STAT-047] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is systematic sampling?

[STAT-048] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the Central Limit Theorem?

[STAT-049] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the Empirical Rule?

[STAT-050] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the Kruskal-Wallis test?

[STAT-051] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the Law of Large Numbers?

[STAT-052] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the Mann-Whitney U test?

[STAT-053] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between a sample and population variance?

[STAT-054] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between covariance and correlation?

[STAT-055] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between descriptive and inferential statistics?

[STAT-056] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between one-tailed and two-tailed tests?

[STAT-057] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the interquartile range?

[STAT-058] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the multiple comparisons problem?

[STAT-059] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is variance of a random variable?

[STAT-060] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is weighted mean?

[STAT-061] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Does correlation imply causation?

[STAT-062] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do outliers affect mean and standard deviation?

[STAT-063] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do outliers affect median?

[STAT-064] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you detect outliers using IQR?

[STAT-065] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you choose a statistical test for a real business problem?

[STAT-066] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What assumptions are required for parametric tests?

[STAT-067] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When is median preferred over mean?

[STAT-068] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why are squared deviations used in variance?

[STAT-069] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What is geometric mean?

[STAT-070] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What is harmonic mean?

[STAT-071] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What is the Wilcoxon signed-rank test?

TOPIC 2: PROBABILITY AND PROBABILITY DISTRIBUTIONS

Total: **17 Questions** | High Priority: **17** | Medium: **0** | Lower: **0**

[STAT-072] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are mean and variance of a Normal distribution?

[STAT-073] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Bernoulli distribution?

[STAT-074] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Beta distribution?

[STAT-075] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Binomial distribution?

[STAT-076] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Gamma distribution?

[STAT-077] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Log-Normal distribution?

[STAT-078] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Normal distribution?

[STAT-079] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Poisson distribution?

[STAT-080] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a cumulative distribution function?

[STAT-081] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a probability density function?

[STAT-082] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a probability mass function?

[STAT-083] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a uniform distribution?

[STAT-084] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an Exponential distribution?

[STAT-085] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is conditional probability?

[STAT-086] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is probability?

[STAT-087] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is sampling distribution?

[STAT-088] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the standard Normal distribution?

TOPIC 3: HYPOTHESIS TESTING AND STATISTICAL INFERENCE

Total: **20 Questions** | High Priority: **15** | Medium: **5** | Lower: **0**

[STAT-089] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is ANOVA?

[STAT-090] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Type I error?

[STAT-091] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a null hypothesis?

[STAT-092] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a one-sample T-test?

[STAT-093] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a p-value?

[STAT-094] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a paired T-test?

[STAT-095] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an alternative hypothesis?

[STAT-096] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an independent two-sample T-test?

[STAT-097] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is hypothesis testing?

[STAT-098] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is multiple hypothesis testing?

[STAT-099] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is significance level alpha?

[STAT-100] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is statistical significance?

[STAT-101] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is statistical versus practical significance?

[STAT-102] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the Chi-square test?

[STAT-103] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the null hypothesis in ANOVA?

[STAT-104] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What assumptions does a T-test make?

[STAT-105] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When do you use a Chi-square goodness-of-fit test?

[STAT-106] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When do you use a Chi-square test of independence?

[STAT-107] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When do you use a Z-test?

[STAT-108] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When should ANOVA be used?

TOPIC 4: EXPLORATORY DATA ANALYSIS (EDA)

Total: **44 Questions** | High Priority: **10** | Medium: **33** | Lower: **1**

[EDA-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How do you detect data leakage during EDA?

[EDA-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are the first steps when receiving a new dataset?

[EDA-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Exploratory Data Analysis?

[EDA-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a bar chart used for?

[EDA-005] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a box plot used for?

[EDA-006] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a histogram used for?

[EDA-007] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a scatter plot used for?

[EDA-008] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a violin plot used for?

[EDA-009] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is bivariate analysis?

[EDA-010] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is multivariate analysis?

[EDA-011] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you analyze a target variable?

[EDA-012] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you analyze categorical variables?

[EDA-013] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you analyze correlation?

[EDA-014] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you analyze missingness patterns?

[EDA-015] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you analyze numerical distributions?

[EDA-016] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you analyze relationships between two numerical variables?

[EDA-017] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you analyze time-based patterns?

[EDA-018] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you avoid misleading visualizations?

[EDA-019] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you choose the right visualization?

[EDA-020] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you communicate EDA findings to stakeholders?

[EDA-021] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you compare distributions across groups?

[EDA-022] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you convert EDA findings into feature engineering ideas?

[EDA-023] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you create an EDA summary report?

[EDA-024] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you detect anomalies?

[EDA-025] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you detect sampling bias during EDA?

[EDA-026] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you detect skewness?

[EDA-027] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you document data quality issues?

[EDA-028] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you identify class imbalance?

[EDA-029] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you identify data types?

[EDA-030] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you identify missing values?

[EDA-031] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you identify multicollinearity?

[EDA-032] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you identify outliers?

[EDA-033] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you identify suspicious values?

[EDA-034] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you identify trend and seasonality?

[EDA-035] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you inspect dataset shape and schema?

[EDA-036] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you perform univariate analysis?

[EDA-037] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you prioritize data cleaning tasks?

[EDA-038] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you validate business rules in data?

[EDA-039] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you perform EDA before building a machine learning model?

[EDA-040] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you perform EDA for a customer churn dataset?

[EDA-041] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you perform EDA for a sales dataset?

[EDA-042] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What visualizations help detect outliers?

[EDA-043] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why is EDA important before modeling?

[EDA-044] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What are limitations of a correlation matrix?

TOPIC 5: FEATURE ENGINEERING AND FEATURE SELECTION

Total: **49 Questions** | High Priority: **37** | Medium: **10** | Lower: **2**

[FE-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are MCAR, MAR, and MNAR?

[FE-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are domain-specific features?

[FE-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are filter feature selection methods?

[FE-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are lag features?

[FE-005] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are risks of target encoding?

[FE-006] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are rolling window features?

[FE-007] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Min-Max scaling?

[FE-008] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is PCA?

[FE-009] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Recursive Feature Elimination?

[FE-010] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Robust scaling?

[FE-011] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is SelectKBest?

[FE-012] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is binning?

[FE-013] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is clipping?

[FE-014] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is date and time feature engineering?

[FE-015] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is dimensionality reduction?

[FE-016] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is discretization?

[FE-017] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is explained variance?

[FE-018] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is feature engineering?

[FE-019] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is feature extraction?

[FE-020] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is feature hashing?

[FE-021] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is feature importance?

[FE-022] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is feature interaction?

[FE-023] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is feature scaling?

[FE-024] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is feature selection?

[FE-025] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is feature skewness?

[FE-026] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is feature transformation?

[FE-027] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is frequency encoding?

[FE-028] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is label encoding?

[FE-029] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is log transformation?

[FE-030] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is mutual information?

[FE-031] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is normalization?

[FE-032] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is one-hot encoding?

[FE-033] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is ordinal encoding?

[FE-034] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is polynomial feature generation?

[FE-035] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is standardization?

[FE-036] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is target encoding?

[FE-037] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is winsorization?

[FE-038] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How can VIF help feature selection?

[FE-039] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you extract features from timestamps?

[FE-040] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you identify redundant features?

[FE-041] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you monitor feature drift?

[FE-042] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you validate a feature engineering pipeline?

[FE-043] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How would you engineer features for a real business dataset?

[FE-044] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
When is log transformation useful?

[FE-045] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
When should Robust scaling be used?

[FE-046] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
Which algorithms require feature scaling?

[FE-047] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
Why is feature engineering important?

[FE-048] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>
What is Box-Cox transformation?

[FE-049] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>
What is Yeo-Johnson transformation?

TOPIC 6: DATA PREPROCESSING AND DATA LEAKAGE

Total: **11 Questions** | High Priority: **6** | Medium: **5** | Lower: **0**

[FE-050] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What are missingness indicators?

[FE-051] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is KNN imputation?

[FE-052] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is leakage-prone feature engineering?

[FE-053] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is mean imputation?

[FE-054] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is median imputation?

[FE-055] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is mode imputation?

[FE-056] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do pipelines prevent preprocessing leakage?

[FE-057] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you handle missing numerical values?

[FE-058] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you handle outliers during feature engineering?

[FE-059] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you prevent leakage during target encoding?

[FE-060] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
Why should features be standardized before PCA?

PART II
SQL, PYTHON AND DATA ANALYSIS

TOPIC 7: SQL FUNDAMENTALS AND DATABASE CONCEPTS

Total: **115 Questions** | High Priority: **77** | Medium: **38** | Lower: **0**

[SQL-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What are ACID properties?

[SQL-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What are SQL string functions?

[SQL-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What are aggregate functions?

[SQL-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are database normal forms?

[SQL-005] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are date data types?

[SQL-006] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is AVG?

[SQL-007] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is CASE WHEN?

[SQL-008] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is COALESCE?

[SQL-009] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is COMMIT?

[SQL-010] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is CONCAT?

[SQL-011] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is COUNT?

[SQL-012] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is DATE TRUNC?

[SQL-013] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is DISTINCT?

[SQL-014] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is ELT?

[SQL-015] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is ETL?

[SQL-016] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is EXISTS?

[SQL-017] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is FIRST VALUE?

[SQL-018] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is GROUP BY?

[SQL-019] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is HAVING?

[SQL-020] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is LAST VALUE?

[SQL-021] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is LIKE?

[SQL-022] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is LIMIT?

[SQL-023] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is LOWER and UPPER?

[SQL-024] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is MAX?

[SQL-025] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is MIN?

[SQL-026] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is NOT EXISTS?

[SQL-027] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is NTILE?

[SQL-028] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is NULL?

[SQL-029] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is NULLIF?

[SQL-030] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is OFFSET?

[SQL-031] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is ORDER BY?

[SQL-032] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is RANGE BETWEEN?

[SQL-033] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is ROLLBACK?

[SQL-034] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is ROW NUMBER?

[SQL-035] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is ROWS BETWEEN?

[SQL-036] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is SELECT?

[SQL-037] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is SQL?

[SQL-038] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is SUBSTRING?

[SQL-039] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is SUM?

[SQL-040] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is TRIM?

[SQL-041] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is WHERE?

[SQL-042] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Cartesian product?

[SQL-043] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a clustered index?

[SQL-044] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a composite key?

[SQL-045] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a database schema?

[SQL-046] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a deadlock?

[SQL-047] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a dimension table?

[SQL-048] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a fact table?

[SQL-049] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a foreign key?

[SQL-050] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a materialized view?

[SQL-051] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a non-clustered index?

[SQL-052] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a pivot in SQL?

[SQL-053] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a primary key?

[SQL-054] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a recursive CT E?

[SQL-055] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a regular expression in SQL?

[SQL-056] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a snowflake schema?

[SQL-057] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a star schema?

[SQL-058] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a transaction?

[SQL-059] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a unique key?

[SQL-060] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a view?

[SQL-061] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an execution plan?

[SQL-062] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an index?

[SQL-063] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an unpivot?

[SQL-064] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is conditional aggregation?

[SQL-065] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is denormalization?

[SQL-066] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is logical query processing order?

[SQL-067] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is partitioning?

[SQL-068] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is predicate pushdown?

[SQL-069] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is running total?

[SQL-070] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is table clustering?

[SQL-071] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between COUNT(*) and COUNT(column)?

[SQL-072] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between DDL and DML?

[SQL-073] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between DELETE, DROP, and TRUNCATE?

[SQL-074] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between EXISTS and IN?

[SQL-075] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between WHERE and HAVING?

[SQL-076] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between a view and a materialized view?

[SQL-077] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is transaction isolation?

[SQL-078] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do indexes improve query performance?

[SQL-079] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you add or subtract dates?

[SQL-080] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you avoid double counting revenue?

[SQL-081] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate a moving average in SQL?

[SQL-082] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate conversion rate?

[SQL-083] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate cumulative percentages?

[SQL-084] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate customer lifetime value components?

[SQL-085] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate date differences?

[SQL-086] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate median in SQL?

[SQL-087] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate percentages in SQL?

[SQL-088] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate percentiles in SQL?

[SQL-089] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate rolling 7-day active users?

[SQL-090] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate sessionization in SQL?

[SQL-091] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you debug an incorrect aggregation?

[SQL-092] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you detect anomalies using SQL?

[SQL-093] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you extract text using regex?

[SQL-094] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you extract year, month, and day?

[SQL-095] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you find duplicate records?

[SQL-096] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you find records present in one table but not another?

[SQL-097] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you find the nth highest value?

[SQL-098] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you find the second highest salary?

[SQL-099] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you find top N records per group?

[SQL-100] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you find users active on consecutive days?

[SQL-101] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you group data by week or month?

[SQL-102] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle NULL values in analytical queries?

[SQL-103] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you identify consecutive dates?

[SQL-104] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you identify reactivated users?

[SQL-105] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you identify the first and latest event per user?

[SQL-106] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you optimize a slow SQL query?

[SQL-107] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you remove duplicates?

[SQL-108] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you validate SQL query results?

[SQL-109] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you write SQL for an AIB test analysis?

[SQL-110] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does AVG handle NULL values?

[SQL-111] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does NULL behave in SQL comparisons?

[SQL-112] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you explain SQL query execution order?

[SQL-113] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you solve a complex SQL interview problem step by step?

[SQL-114] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When can indexes hurt performance?

[SQL-115] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why should SELECT * be avoided in large queries?

TOPIC 8: SQL JOINS, SUBQUERIES AND SET OPERATIONS

Total: **19 Questions** | High Priority: **15** | Medium: **4** | Lower: **0**

[SQL-116] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is CROSS JOIN?

[SQL-117] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is EXCEPT or MINUS?

[SQL-118] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is FULL OUTER JOIN?

[SQL-119] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is INNER JOIN?

[SQL-120] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is INTERSECT?

[SQL-121] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is LEFT JOIN?

[SQL-122] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is RIGHT JOIN?

[SQL-123] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is SELF JOIN?

[SQL-124] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is UNION ALL?

[SQL-125] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is UNION?

[SQL-126] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a correlated subquery?

[SQL-127] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a subquery?

[SQL-128] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between INNER and LEFT JOIN?

[SQL-129] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between UNION and UNION ALL?

[SQL-130] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between a subquery and a CTE?

[SQL-131] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle many-to-many joins?

[SQL-132] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you prevent duplicate rows after joins?

[SQL-133] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you optimize a query joining very large tables?

[SQL-134] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What happens when JOIN keys contain duplicates?

TOPIC 9: SQL AGGREGATIONS, CTES AND WINDOW FUNCTIONS

Total: **17 Questions** | High Priority: **12** | Medium: **5** | Lower: **0**

[SQL-135] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How do CTEs affect readability and performance?

[SQL-136] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How do window functions differ from GROUP BY?

[SQL-137] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are wildcard characters?

[SQL-138] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is DENSE RANK?

[SQL-139] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is LAG?

[SQL-140] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is LEAD?

[SQL-141] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is ORDER BY inside OVER?

[SQL-142] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is PARTITION BY?

[SQL-143] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is RANK?

[SQL-144] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a CTE?

[SQL-145] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a frame clause in a window function?

[SQL-146] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a window function?

[SQL-147] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate month-over-month growth?

[SQL-148] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate rank within each category?

[SQL-149] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate year-over-year growth?

[SQL-150] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you find customers with no orders?

[SQL-151] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you use CASE WHEN with aggregation?

TOPIC 10: ADVANCED SQL AND BUSINESS ANALYTICS PROBLEMS

Total: **7 Questions** | High Priority: **0** | Medium: **7** | Lower: **0**

[SQL-152] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you build a cohort retention table?

[SQL-153] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate DAU, WAU, and MAU?

[SQL-154] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate churn using SQL?

[SQL-155] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate retention cohorts?

[SQL-156] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you find gaps and islands?

[SQL-157] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you write a funnel analysis query?

[SQL-158] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you design SQL for a business KPI dashboard?

TOPIC 11: PYTHON FUNDAMENTALS AND DATA STRUCTURES

Total: **58 Questions** | High Priority: **52** | Medium: **6** | Lower: **0**

[PY-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are *args and **kwargs?

[PY-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are Python's main built-in data types?

[PY-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are risks of unpickling untrusted data?

[PY-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are unit tests?

[PY-005] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is JSON serialization?

[PY-006] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is LEGB scope?

[PY-007] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a closure?

[PY-008] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a context manager?

[PY-009] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a dictionary?

[PY-010] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a function?

[PY-011] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a lambda function?

[PY-012] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a static method?

[PY-013] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a virtual environment?

[PY-014] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is abstraction?

[PY-015] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an iterable?

[PY-016] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is array shape?

[PY-017] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is boolean indexing?

[PY-018] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is broadcasting?

[PY-019] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is deep copy?

[PY-020] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is dictionary comprehension?

[PY-021] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is encapsulation?

[PY-022] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is exception handling?

[PY-023] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is filter?

[PY-024] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is flatten versus ravel?

[PY-025] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is init ?

[PY-026] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is list comprehension?

[PY-027] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is logging?

[PY-028] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is map?

[PY-029] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is mutability?

[PY-030] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is pickle?

[PY-031] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is pip?

[PY-032] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is polymorphism?

[PY-033] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is pytest?

[PY-034] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is random seed?

[PY-035] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is reduce?

[PY-036] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is reference counting?

[PY-037] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is requirements.txt?

[PY-038] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is reshape?

[PY-039] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is serialization?

[PY-040] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is shallow copy?

[PY-041] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is slicing?

[PY-042] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is str ?

[PY-043] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the Global Interpreter Lock?

[PY-044] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between == and is?

[PY-045] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between a list and tuple?

[PY-046] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between a set and list?

[PY-047] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between break and return?

[PY-048] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between pass and continue?

[PY-049] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the with statement?

[PY-050] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is try-except-finally?

[PY-051] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is type hinting?

[PY-052] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is variable scope?

[PY-053] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you debug Python code?

[PY-054] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you profile Python performance?

[PY-055] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you read large files efficiently?

[PY-056] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you structure a data science Python project?

[PY-057] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you write efficient Python code?

[PY-058] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you optimize a slow Python data pipeline?

TOPIC 12: PYTHON OOP, ITERATORS, GENERATORS AND DECORATORS

Total: **12 Questions** | High Priority: **10** | Medium: **2** | Lower: **0**

[PY-059] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a class method?

[PY-060] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a class?

[PY-061] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a dataclass?

[PY-062] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a decorator?

[PY-063] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a generator expression?

[PY-064] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a generator?

[PY-065] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an iterator?

[PY-066] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an object?

[PY-067] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is inheritance?

[PY-068] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is yield?

[PY-069] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Which Python objects are mutable?

[PY-070] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why is NumPy faster than Python loops?

TOPIC 13: PYTHON PERFORMANCE, CONCURRENCY AND MEMORY

Total: **8 Questions** | High Priority: **6** | Medium: **1** | Lower: **1**

[PY-071] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Big-O notation?

[PY-072] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Python garbage collection?

[PY-073] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is multiprocessing?

[PY-074] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is multithreading?

[PY-075] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the average lookup complexity of a dictionary?

[PY-076] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the time complexity of list lookup?

[PY-077] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When is multiprocessing preferred?

[PY-078] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

How do you handle memory limitations in Python?

TOPIC 14: NUMPY AND VECTORIZED COMPUTATION

Total: **5 Questions** | High Priority: **4** | Medium: **1** | Lower: **0**

[PY-079] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is NumPy axis?

[PY-080] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is NumPy?

[PY-081] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a NumPy ndarray?

[PY-082] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is vectorization?

[PY-083] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you explain vectorization to an interviewer?

TOPIC 15: PANDAS AND DATA MANIPULATION

Total: **57 Questions** | High Priority: **33** | Medium: **24** | Lower: **0**

[PD-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are categorical dtypes?

[PD-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are inner, left, right, and outer merges?

[PD-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is MultiIndex?

[PD-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Pandas?

[PD-005] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is SettingWithCopyWarning?

[PD-006] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a DataFrame?

[PD-007] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Series?

[PD-008] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a pivot table?

[PD-009] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is aggregation?

[PD-010] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is apply?

[PD-011] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is astype?

[PD-012] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is describe?

[PD-013] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is dropna?

[PD-014] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is expanding?

[PD-015] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is fillna?

[PD-016] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is groupby?

[PD-017] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is head?

[PD-018] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is iloc?

[PD-019] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is info?

[PD-020] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is isna?

[PD-021] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is loc?

[PD-022] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is melt?

[PD-023] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is merge?

[PD-024] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is pd.to_datetime?

[PD-025] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is pivot?

[PD-026] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is read_parquet?

[PD-027] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is rolling?

[PD-028] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is shift?

[PD-029] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is stack and unstack?

[PD-030] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between apply and vectorization?

[PD-031] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between loc and iloc?

[PD-032] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between merge and join?

[PD-033] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is transform?

[PD-034] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you avoid chained assignment?

[PD-035] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate correlations?

[PD-036] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you calculate rolling averages?

[PD-037] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you change data types?

[PD-038] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you compare two DataFrames?

[PD-039] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you create a DataFrame?

[PD-040] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you create lag features?

[PD-041] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you extract year and month from dates?

[PD-042] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you filter rows?

[PD-043] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you find outliers in a DataFrame?

[PD-044] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle schema changes?

[PD-045] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you optimize Pandas memory usage?

[PD-046] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you process large datasets with chunks?

[PD-047] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you read CSV files?

[PD-048] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you rename columns?

[PD-049] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you resample time series?

[PD-050] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you reset an index?

[PD-051] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you select columns?

[PD-052] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you sort a DataFrame?

[PD-053] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you validate a DataFrame after transformations?

[PD-054] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you work with datetime columns?

[PD-055] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What alternatives exist to iterrows?

[PD-056] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When should you use NumPy instead of Pandas?

[PD-057] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why is iterrows slow?

TOPIC 16: DATA CLEANING AND ANALYTICAL PIPELINES

Total: **3 Questions** | High Priority: **0** | Medium: **3** | Lower: **0**

[PD-058] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you detect duplicates?

[PD-059] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you debug a slow Pandas pipeline?

[PD-060] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What causes duplicate rows after a merge?

PART III
MACHINE LEARNING

TOPIC 17: SUPERVISED AND UNSUPERVISED LEARNING

Total: **31 Questions** | High Priority: **20** | Medium: **9** | Lower: **2**

[ML-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How do pipelines prevent data leakage?

[ML-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How does cross-validation work for regression?

[ML-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are features and target variables?

[ML-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is MAE?

[ML-005] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is MSE?

[ML-006] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is RMSE?

[ML-007] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a decision boundary?

[ML-008] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a loss function?

[ML-009] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an objective function?

[ML-010] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is binary classification?

[ML-011] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is empirical risk minimization?

[ML-012] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is generalization?

[ML-013] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is multiclass classification?

[ML-014] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is multilabel classification?

[ML-015] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is overfitting?

[ML-016] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is regularization?

[ML-017] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between parametric and non-parametric models?

[ML-018] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between regression and classification?

[ML-019] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between supervised learning and unsupervised learning?

[ML-020] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is underfitting?

[ML-021] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you encode categorical variables?

[ML-022] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle missing values before regression?

[ML-023] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you implement Linear Regression using scikit-learn?

[ML-024] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you interpret residuals?

[ML-025] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you split data for regression?

[ML-026] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What common mistakes occur when applying this technique in production?

[ML-027] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What metrics evaluate regression models?

[ML-028] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When is MAE preferred over RMSE?

[ML-029] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When should features be standardized?

[ML-030] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

In a real-world project, how would you explain this concept to a non-technical stakeholder?

[ML-031] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What is MAPE and what are its limitations?

TOPIC 18: LINEAR AND LOGISTIC REGRESSION

Total: **35 Questions** | High Priority: **24** | Medium: **11** | Lower: **0**

[ML-032] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How can multicollinearity be detected?

[ML-033] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are odds and log-odds?

[ML-034] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What does a predicted probability represent?

[ML-035] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Adjusted R-squared?

[ML-036] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Linear Regression?

[ML-037] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Logistic Regression?

[ML-038] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Ordinary Least Squares?

[ML-039] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is R-squared?

[ML-040] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is VIF?

[ML-041] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a logistic decision boundary?

[ML-042] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is binary cross-entropy?

[ML-043] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is gradient descent?

[ML-044] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is heteroscedasticity?

[ML-045] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is homoscedasticity?

[ML-046] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is maximum likelihood estimation?

[ML-047] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is mini-batch gradient descent?

[ML-048] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is multicollinearity?

[ML-049] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is multinomial Logistic Regression?

[ML-050] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is multiple Linear Regression?

[ML-051] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is one-vs-rest classification?

[ML-052] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is simple Linear Regression?

[ML-053] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is stochastic gradient descent?

[ML-054] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is the Mean Squared Error loss?

[ML-055] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is the sigmoid function?

[ML-056] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How are Logistic Regression coefficients interpreted?

[ML-057] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do outliers affect Linear Regression?

[ML-058] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do residual plots help?

[ML-059] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you handle class imbalance?

[ML-060] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you handle non-linear relationships?

[ML-061] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How does changing the classification threshold affect predictions?

[ML-062] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How does regularization work in Logistic Regression?

[ML-063] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
What assumptions does Linear Regression make?

[ML-064] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
What loss function is used?

[ML-065] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why are squared errors used?

[ML-066] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why is Logistic Regression used for classification?

TOPIC 19: REGULARIZATION: RIDGE, LASSO AND ELASTIC NET

Total: **24 Questions** | High Priority: **10** | Medium: **14** | Lower: **0**

[ML-067] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How do you interpret features selected by Lasso?

[ML-068] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How does regularization affect bias and variance?

[ML-069] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Should features be scaled before Ridge and Lasso?

[ML-070] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Elastic Net?

[ML-071] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Lasso Regression?

[ML-072] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is LassoCV?

[ML-073] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Ridge Regression?

[ML-074] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is RidgeCV?

[ML-075] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between L1 and L2 regularization?

[ML-076] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the regularization parameter?

[ML-077] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you compare Linear, Ridge, and Lasso?

[ML-078] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you implement Lasso Regression?

[ML-079] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you implement Ridge Regression?

[ML-080] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you select the best regularization strength?

[ML-081] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you tune alpha?

[ML-082] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How does Lasso perform feature selection?

[ML-083] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How does Ridge affect coefficients?

[ML-084] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
What happens when alpha is too high?

[ML-085] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
What happens when regularization is very high?

[ML-086] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
What happens when regularization is zero?

[ML-087] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
When would you prefer Ridge over Lasso?

[ML-088] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
Why is Elastic Net useful for correlated features?

[ML-089] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
Why is regularization needed?

[ML-090] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
Why should scaling be inside a pipeline?

TOPIC 20: NAIVE BAYES AND K-NEAREST NEIGHBORS

Total: **25 Questions** | High Priority: **12** | Medium: **11** | Lower: **2**

[ML-091] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What are prior, likelihood, and posterior probabilities?

[ML-092] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is Bayes theorem?

[ML-093] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Bernoulli Naive Bayes?

[ML-094] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Euclidean distance?

[ML-095] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Gaussian Naive Bayes?

[ML-096] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is KNN?

[ML-097] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Laplace smoothing?

[ML-098] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Manhattan distance?

[ML-099] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Multinomial Naive Bayes?

[ML-100] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Naive Bayes?

[ML-101] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the curse of dimensionality?

[ML-102] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the zero-frequency problem?

[ML-103] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How can KNN handle imbalanced datasets?

[ML-104] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you choose K?

[ML-105] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does KNN make predictions?

[ML-106] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does KNN perform regression?

[ML-107] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What conditional independence assumption does it make?

[ML-108] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What distance metrics can KNN use?

[ML-109] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What happens when K is too large?

[ML-110] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What happens when K is too small?

[ML-111] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why is KNN called a lazy learner?

[ML-112] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why is feature scaling important?

[ML-113] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why is it called naive?

[ML-114] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What are KNN's computational limitations?

[ML-115] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What are Naive Bayes limitations?

TOPIC 21: DECISION TREES

Total: **22 Questions** | High Priority: **11** | Medium: **10** | Lower: **1**

[ML-116] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are Decision Tree advantages?

[ML-117] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are pre-pruning and post-pruning?

[ML-118] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are root, internal, and leaf nodes?

[ML-119] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Gini impurity?

[ML-120] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Decision Tree?

[ML-121] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is entropy?

[ML-122] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is information gain?

[ML-123] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is maximum depth?

[ML-124] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is minimum samples leaf?

[ML-125] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is minimum samples split?

[ML-126] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between Gini and entropy?

[ML-127] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Can Regression Trees extrapolate?

[ML-128] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you prune a Regression Tree?

[ML-129] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does a Decision Tree choose a split?

[ML-130] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does a Regression Tree differ from a Classification Tree?

[ML-131] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does a Regression Tree make predictions?

[ML-132] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does tree depth affect performance?

[ML-133] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How is feature importance calculated?

[ML-134] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What causes Decision Trees to overfit?

[ML-135] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What split criterion is used?

[ML-136] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why do Regression Trees produce piecewise constant predictions?

[ML-137] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What are Decision Tree disadvantages?

TOPIC 22: ENSEMBLE LEARNING, BAGGING AND BOOSTING

Total: **14 Questions** | High Priority: **10** | Medium: **4** | Lower: **0**

[ML-138] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a strong learner?

[ML-139] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a weak learner?

[ML-140] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is bagging?

[ML-141] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is blending?

[ML-142] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is boosting?

[ML-143] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is bootstrap sampling?

[ML-144] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is ensemble learning?

[ML-145] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is out-of-bag evaluation?

[ML-146] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is stacking?

[ML-147] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between bagging and boosting?

[ML-148] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does bagging reduce variance?

[ML-149] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does boosting reduce bias?

[ML-150] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why can boosting be sensitive to noisy data?

[ML-151] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why can combining models improve performance?

TOPIC 23: RANDOM FOREST

Total: **13 Questions** | High Priority: **4** | Medium: **8** | Lower: **1**

[ML-152] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are its advantages?

[ML-153] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Random Forest?

[ML-154] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is out-of-bag error?

[ML-155] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is permutation importance?

[ML-156] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Can Random Forest overfit?

[ML-157] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does Random Forest handle imbalanced classes?

[ML-158] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does Random Forest reduce correlation between trees?

[ML-159] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does Random Forest use bagging?

[ML-160] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does max_depth affect the model?

[ML-161] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does max_features affect the model?

[ML-162] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What hyperparameters are important?

[ML-163] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why are random feature subsets used?

[ML-164] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What are its disadvantages?

TOPIC 24: GRADIENT BOOSTING, ADABOOST AND XGBOOST

Total: **34 Questions** | High Priority: **15** | Medium: **19** | Lower: **0**

[ML-165] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How does subsampling reduce overfitting?

[ML-166] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are gamma, lambda, and alpha?

[ML-167] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are subsample and colsample_bytree?

[ML-168] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is AdaBoost?

[ML-169] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Gradient Boosting?

[ML-170] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is XGBoost feature importance?

[ML-171] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is XGBoost?

[ML-172] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a decision stump?

[ML-173] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is early stopping?

[ML-174] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is max_depth?

[ML-175] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is min_child_weight?

[ML-176] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is shrinkage?

[ML-177] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between XGBoost and Random Forest?

[ML-178] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the learning-rate versus estimators tradeoff?

[ML-179] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the role of gradients and Hessians?

[ML-180] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

Can XGBoost model non-linear relationships?

[ML-181] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How are sample weights updated?

[ML-182] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How can boosting models be regularized?

[ML-183] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you compare XGBoost with Random Forest for regression?

[ML-184] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you evaluate an XGBoost Regressor?

[ML-185] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you tune an XGBoost Regressor?

[ML-186] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How does AdaBoost train models sequentially?

[ML-187] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How does Gradient Boosting differ from AdaBoost?

[ML-188] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How does XGBoost differ from traditional Gradient Boosting?

[ML-189] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How does XGBoost handle missing values?

[ML-190] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How does early stopping work?

[ML-191] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How does max_depth affect regression?

[ML-192] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How does scale_pos_weight handle class imbalance?

[ML-193] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

What causes boosting to overfit?

[ML-194] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What objectives can XGBoost use for regression?

[ML-195] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What regularization does XGBoost use?

[ML-196] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why are misclassified samples given more weight?

[ML-197] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why is AdaBoost sensitive to outliers?

[ML-198] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why is XGBoost computationally efficient?

TOPIC 25: CLUSTERING: K-MEANS, HIERARCHICAL CLUSTERING AND DBSCAN

Total: **45 Questions** | High Priority: **19** | Medium: **23** | Lower: **3**

[ML-199] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is DBSCAN?

[ML-200] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Hierarchical Clustering?

[ML-201] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is K-Means?

[ML-202] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Ward linkage?

[ML-203] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a border point?

[ML-204] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a centroid?

[ML-205] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a core point?

[ML-206] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a dendrogram?

[ML-207] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a k-distance plot?

[ML-208] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a noise point?

[ML-209] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is agglomerative clustering?

[ML-210] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is average linkage?

[ML-211] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is complete linkage?

[ML-212] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is divisive clustering?

[ML-213] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is inertia?

[ML-214] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is single linkage?

[ML-215] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the Silhouette Score?

[ML-216] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the elbow method?

[ML-217] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the range of the score?

[ML-218] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How can PCA help clustering?

[ML-219] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do outliers affect K-Means?

[ML-220] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you choose between K-Means and DBSCAN?

[ML-221] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you choose clusters from a dendrogram?

[ML-222] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you choose epsilon?

[ML-223] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you evaluate whether clusters are useful?

[ML-224] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you explain clusters to stakeholders?

[ML-225] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle categorical variables?

[ML-226] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle outliers before clustering?

[ML-227] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How does DBSCAN discover clusters?

[ML-228] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How does DBSCAN handle outliers?

[ML-229] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How does K-Means work step by step?

[ML-230] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How is it used to choose K?

[ML-231] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you cluster customers for segmentation?

[ML-232] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you preprocess customer data for clustering?

[ML-233] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

What assumptions does K-Means make?

[ML-234] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

What do epsilon and min_samples represent?

[ML-235] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

What does a high score indicate?

[ML-236] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

What does a negative score indicate?

[ML-237] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

What objective function does K-Means minimize?

[ML-238] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
Why can DBSCAN find arbitrary shapes?

[ML-239] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
Why can K-Means converge to a local optimum?

[ML-240] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
Why does DBSCAN struggle with varying density?

[ML-241] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>
What are Hierarchical Clustering limitations?

[ML-242] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>
What are K-Means limitations?

[ML-243] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>
What are its limitations?

TOPIC 26: BIAS-VARIANCE TRADEOFF AND GENERALIZATION

Total: **12 Questions** | High Priority: **6** | Medium: **6** | Lower: **0**

[ML-244] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
How do learning curves diagnose bias and variance?

[ML-245] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is bias?

[ML-246] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is high bias?

[ML-247] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is high variance?

[ML-248] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is the bias-variance tradeoff?

[ML-249] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is variance?

[ML-250] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How can you reduce high bias?

[ML-251] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How can you reduce high variance?

[ML-252] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does bagging affect variance?

[ML-253] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does boosting affect bias?

[ML-254] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does regularization affect it?

[ML-255] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why does the tradeoff occur?

PART IV
MODEL EVALUATION AND PRODUCTION ML

TOPIC 27: CLASSIFICATION METRICS AND CONFUSION MATRIX

Total: **33 Questions** | High Priority: **28** | Medium: **5** | Lower: **0**

[MLE-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How do you detect overfitting from training and validation metrics?

[MLE-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How do you detect underfitting?

[MLE-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are false negatives?

[MLE-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are false positives?

[MLE-005] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are true negatives?

[MLE-006] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are true positives?

[MLE-007] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is McNemar's test?

[MLE-008] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Median Absolute Error?

[MLE-009] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a confusion matrix?

[MLE-010] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a test set?

[MLE-011] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a validation curve?

[MLE-012] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a validation set?

[MLE-013] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is accuracy?

[MLE-014] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is baseline model performance?

[MLE-015] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is bootstrapped model evaluation?

[MLE-016] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is class imbalance?

[MLE-017] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is learning curve analysis?

[MLE-018] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is log loss?

[MLE-019] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is macro averaging?

[MLE-020] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is micro averaging?

[MLE-021] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is model robustness?

[MLE-022] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is model stability?

[MLE-023] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is multiclass cross-entropy?

[MLE-024] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is sensitivity?

[MLE-025] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is specificity?

[MLE-026] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the precision-recall tradeoff?

[MLE-027] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is train-test contamination?

[MLE-028] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is weighted averaging?

[MLE-029] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do confidence intervals apply to model metrics?

[MLE-030] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you compare two classifiers statistically?

[MLE-031] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you choose the correct metric for a business problem?

[MLE-032] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When can accuracy be misleading?

[MLE-033] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why should models be compared using the same validation strategy?

TOPIC 28: REGRESSION METRICS

Total: **2 Questions** | High Priority: **1** | Medium: **0** | Lower: **1**

[MLE-034] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is MAPE?

[MLE-035] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What are limitations of MAPE?

TOPIC 29: ROC-AUC, PR-AUC AND THRESHOLD OPTIMIZATION

Total: **8 Questions** | High Priority: **5** | Medium: **3** | Lower: **0**

[MLE-036] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How does threshold selection affect precision and recall?

[MLE-037] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is AUC-ROC?

[MLE-038] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Precision-Recall curve?

[MLE-039] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a classification threshold?

[MLE-040] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an ROC curve?

[MLE-041] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you select an optimal classification threshold?

[MLE-042] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What does an AUC of 0.5 mean?

[MLE-043] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When is a Precision-Recall curve preferred over ROC?

TOPIC 30: IMBALANCED CLASSIFICATION

Total: **2 Questions** | High Priority: **1** | Medium: **1** | Lower: **0**

[MLE-044] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is F-beta score?

[MLE-045] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you evaluate an imbalanced classification problem?

TOPIC 31: CROSS-VALIDATION AND HYPERPARAMETER TUNING

Total: **11 Questions** | High Priority: **10** | Medium: **1** | Lower: **0**

[MLE-046] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Bayesian optimization for hyperparameters?

[MLE-047] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Grid Search?

[MLE-048] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is K-Fold cross-validation?

[MLE-049] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Leave-One-Out cross-validation?

[MLE-050] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Random Search?

[MLE-051] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Repeated K-Fold?

[MLE-052] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Stratified K-Fold?

[MLE-053] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is cross-validation?

[MLE-054] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is hyperparameter tuning?

[MLE-055] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is nested cross-validation?

[MLE-056] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why should the test set not be used for hyperparameter tuning?

TOPIC 32: CALIBRATION, ROBUSTNESS AND MODEL SELECTION

Total: **4 Questions** | High Priority: **3** | Medium: **0** | Lower: **1**

[MLE-057] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a calibration curve?

[MLE-058] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is calibration?

[MLE-059] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is model selection?

[MLE-060] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What is Brier score?

TOPIC 33: ML PIPELINES AND DATA LEAKAGE PREVENTION

Total: **2 Questions** | High Priority: **2** | Medium: **0** | Lower: **0**

[MLE-061] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is data leakage?

[MLE-062] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is target leakage?

PART V
DEEP LEARNING

TOPIC 34: NEURAL NETWORKS AND BACKPROPAGATION

Total: **45 Questions** | High Priority: **39** | Medium: **5** | Lower: **1**

[DL-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How does gradient descent train a neural network?

[DL-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are many-to-many architectures?

[DL-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are many-to-one architectures?

[DL-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are weights and biases?

[DL-005] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What causes overfitting in CNNs?

[DL-006] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is ReLU?

[DL-007] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a CNN?

[DL-008] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a convolutional filter or kernel?

[DL-009] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a feature map?

[DL-010] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a hidden layer?

[DL-011] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a hidden state?

[DL-012] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a learning rate?

[DL-013] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a receptive field?

[DL-014] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an Artificial Neural Network?

[DL-015] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an RNN?

[DL-016] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an activation function?

[DL-017] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an artificial neuron?

[DL-018] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an epoch?

[DL-019] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is average pooling?

[DL-020] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is backpropagation through time?

[DL-021] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is backpropagation?

[DL-022] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is batch normalization?

[DL-023] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is batch size?

[DL-024] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is convolution?

[DL-025] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is data augmentation?

[DL-026] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is dropout?

[DL-027] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is forward propagation?

[DL-028] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is max pooling?

[DL-029] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is padding?

[DL-030] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is parameter sharing?

[DL-031] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is pooling?

[DL-032] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is sequence-to-sequence learning?

[DL-033] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is sigmoid?

[DL-034] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is stride?

[DL-035] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is tanh?

[DL-036] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the exploding gradient problem?

[DL-037] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the vanishing gradient problem?

[DL-038] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is transfer learning in CNNs?

[DL-039] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is weight initialization?

[DL-040] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What applications commonly use RNNs?

[DL-041] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why are CNNs effective for images?

[DL-042] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why are RNNs useful for sequential data?

[DL-043] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why are non-linear activation functions required?

[DL-044] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why do vanilla RNNs suffer from vanishing gradients?

[DL-045] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What are the limitations of vanilla RNNs?

TOPIC 35: RNNS, LSTM, GRU AND BILSTM

Total: **32 Questions** | High Priority: **10** | Medium: **16** | Lower: **6**

[DL-046] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are GRU advantages?

[DL-047] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are update and reset gates?

[DL-048] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a BiLSTM?

[DL-049] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a GRU?

[DL-050] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a cell state?

[DL-051] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an LSTM?

[DL-052] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between hidden state and cell state?

[DL-053] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the forget gate?

[DL-054] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the input gate?

[DL-055] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the output gate?

[DL-056] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How can dropout be applied to recurrent networks?

[DL-057] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you choose between GRU and LSTM?

[DL-058] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does BiLSTM compare with unidirectional LSTM?

[DL-059] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does GRU differ from LSTM?

[DL-060] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does LSTM address vanishing gradients?

[DL-061] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does LSTM compare with GRU?

[DL-062] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does LSTM compare with vanilla RNN?

[DL-063] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does stacking recurrent layers increase capacity?

[DL-064] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What does the backward LSTM capture?

[DL-065] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What does the forward LSTM capture?

[DL-066] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When is BiLSTM unsuitable?

[DL-067] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When might GRU train faster than LSTM?

[DL-068] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When should you use LSTM?

[DL-069] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why does GRU have fewer parameters?

[DL-070] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why does bidirectional processing help?

[DL-071] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why was LSTM introduced?

[DL-072] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What are GRU limitations?

[DL-073] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What are LSTM limitations?

[DL-074] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What are the advantages of Deep RNNs?

[DL-075] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What are the disadvantages of Deep RNNs?

[DL-076] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What is a Deep RNN?

[DL-077] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What training challenges occur in Deep RNNs?

TOPIC 36: ATTENTION AND TRANSFORMER ARCHITECTURE

Total: **22 Questions** | High Priority: **16** | Medium: **6** | Lower: **0**

[DL-078] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are Query, Key, and Value?

[DL-079] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is BERT?

[DL-080] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is GPT?

[DL-081] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Transformer architecture?

[DL-082] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a decoder?

[DL-083] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an encoder?

[DL-084] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is cross-attention?

[DL-085] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is fine-tuning?

[DL-086] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is layer normalization?

[DL-087] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is masked self-attention?

[DL-088] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is multi-head attention?

[DL-089] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is pretraining?

[DL-090] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is scaled dot-product attention?

[DL-091] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is self-attention?

[DL-092] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is the feed-forward network in a Transformer block?

[DL-093] ★★★ PRIORITY: HIGH | VIEW ANSWER >>
What is the quadratic complexity problem?

[DL-094] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do Transformers differ from LSTMs?

[DL-095] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do Transformers differ from RNNs?

[DL-096] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do modern architectures handle long contexts?

[DL-097] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
Why are positional encodings needed?

[DL-098] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
Why is standard attention computationally expensive?

[DL-099] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
Why were Transformers introduced?

TOPIC 37: EMBEDDINGS AND REPRESENTATION LEARNING

Total: **1 Questions** | High Priority: **1** | Medium: **0** | Lower: **0**

[DL-100] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are embeddings?

PART VI
GENERATIVE AI AND LLMs

TOPIC 38: GENERATIVE AI AND LLM FUNDAMENTALS

Total: **29 Questions** | High Priority: **23** | Medium: **6** | Lower: **0**

[LLM-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Explain self-attention.?

[LLM-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are Query, Key, and Value in attention?

[LLM-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are function/tool calls?

[LLM-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are system, human, and AI messages?

[LLM-005] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is DPO?

[LLM-006] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is JSON mode?

[LLM-007] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is RLHF?

[LLM-008] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a foundation model?

[LLM-009] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a system prompt?

[LLM-010] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a transformer?

[LLM-011] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is alignment?

[LLM-012] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an LLM?

[LLM-013] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is beam search and when is it useful?

[LLM-014] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is context engineering?

[LLM-015] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is deterministic generation?

[LLM-016] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is hallucination in LLMs?

[LLM-017] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is instruction tuning?

[LLM-018] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is model latency?

[LLM-019] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is positional encoding?

[LLM-020] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is prompt engineering?

[LLM-021] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is structured output?

[LLM-022] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is the difference between RLHF and DPO?

[LLM-023] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is throughput in LLM inference?

[LLM-024] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you reduce LLM cost?

[LLM-025] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you reduce LLM latency?

[LLM-026] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you choose an LLM for a production application?

[LLM-027] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What is Generative AI, and how is it different from traditional predictive ML?

[LLM-028] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why are transformers better suited than RNNs for large-scale language modeling?

[LLM-029] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why do LLMs hallucinate?

TOPIC 39: TOKENIZATION, EMBEDDINGS AND CONTEXT WINDOWS

Total: **7 Questions** | High Priority: **6** | Medium: **1** | Lower: **0**

[LLM-030] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How does tokenization work?

[LLM-031] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a context window?

[LLM-032] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a token?

[LLM-033] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is time-to-first-token?

[LLM-034] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is token cost?

[LLM-035] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is tokenization?

[LLM-036] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What happens when the context window is exceeded?

TOPIC 40: PROMPT ENGINEERING AND CONTEXT ENGINEERING

Total: **26 Questions** | High Priority: **15** | Medium: **11** | Lower: **0**

[LLM-037] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How do you defend against prompt injection?

[LLM-038] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is ReAct prompting?

[LLM-039] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is chain-of-thought prompting?

[LLM-040] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is constrained decoding?

[LLM-041] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is context compression?

[LLM-042] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is context prioritization?

[LLM-043] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is few-shot prompting?

[LLM-044] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is indirect prompt injection?

[LLM-045] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is instruction hierarchy?

[LLM-046] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is prompt chaining?

[LLM-047] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is prompt injection?

[LLM-048] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is prompt leakage?

[LLM-049] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is role prompting?

[LLM-050] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is self-consistency?

[LLM-051] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is zero-shot prompting?

[LLM-052] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you debug a prompt that works on some examples but fails on others?

[LLM-053] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you design prompts for reliable classification?

[LLM-054] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you force an LLM to return valid JSON?

[LLM-055] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle ambiguous user queries?

[LLM-056] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you manage very long context?

[LLM-057] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you prevent irrelevant context from degrading an answer?

[LLM-058] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you A/B test prompts?

[LLM-059] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you design a prompt for a tool-using agent?

[LLM-060] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you version prompts in production?

[LLM-061] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Should chain-of-thought be exposed to users?

[LLM-062] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What makes a good prompt?

TOPIC 41: LLM INFERENCE, DECODING AND SAMPLING

Total: **4 Questions** | High Priority: **3** | Medium: **1** | Lower: **0**

[LLM-063] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is greedy decoding?

[LLM-064] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is temperature?

[LLM-065] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is top-k sampling?

[LLM-066] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do temperature, top-k, and top-p differ?

TOPIC 42: FINE-TUNING, PEFT, LORA AND QLORA

Total: **34 Questions** | High Priority: **18** | Medium: **16** | Lower: **0**

[LLM-067] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How do you detect data leakage?

[LLM-068] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How do you prevent overfitting during fine-tuning?

[LLM-069] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are LoRA adapters?

[LLM-070] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is LoRA?

[LLM-071] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is PEFT?

[LLM-072] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is QLoRA?

[LLM-073] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is adapter merging?

[LLM-074] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is catastrophic forgetting?

[LLM-075] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is full fine-tuning?

[LLM-076] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is gradient accumulation?

[LLM-077] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is instruction fine-tuning?

[LLM-078] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is learning rate in fine-tuning?

[LLM-079] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is parameter-efficient fine-tuning?

[LLM-080] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is rank in LoRA?

[LLM-081] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is supervised fine-tuning?

[LLM-082] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is train/validation/test splitting for fine-tuning?

[LLM-083] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is warmup?

[LLM-084] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is weight decay?

[LLM-085] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

Can RAG and fine-tuning be used together?

[LLM-086] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you choose LoRA rank?

[LLM-087] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you choose between multiple adapters?

[LLM-088] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you deploy a LoRA adapter?

[LLM-089] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you evaluate a fine-tuned model?

[LLM-090] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you prepare a fine-tuning dataset?

[LLM-091] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you remove duplicates from training data?

[LLM-092] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you version fine-tuned models?

[LLM-093] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How does LoRA work?

[LLM-094] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

How much data is needed for fine-tuning?

[LLM-095] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

What happens if the fine-tuned model performs worse than the base model?

[LLM-096] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

What makes a high-quality instruction dataset?

[LLM-097] ★★★ PRIORITY: MEDIUM | VIEW ANSWER >>

What metrics should be used for fine-tuning?

[LLM-098] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When should you fine-tune an LLM?

[LLM-099] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why can fine-tuning improve style but hurt factuality?

[LLM-100] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why does QLoRA reduce memory requirements?

TOPIC 43: RLHF, DPO AND MODEL ALIGNMENT

Total: **12 Questions** | High Priority: **9** | Medium: **3** | Lower: **0**

[LLM-101] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are the stages of RLHF?

[LLM-102] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Constitutional AI?

[LLM-103] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is PPO?

[LLM-104] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a reward model?

[LLM-105] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is alignment tax?

[LLM-106] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is preference data?

[LLM-107] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is rejection sampling?

[LLM-108] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is reward hacking?

[LLM-109] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is specification gaming?

[LLM-110] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you evaluate safety after fine-tuning?

[LLM-111] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you prevent a fine-tuning dataset from teaching unsafe behavior?

[LLM-112] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does DPO work conceptually?

PART VII
RAG AND INFORMATION RETRIEVAL

TOPIC 44: RAG FUNDAMENTALS AND ARCHITECTURE

Total: **19 Questions** | High Priority: **15** | Medium: **4** | Lower: **0**

[RAG-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Explain a RAG pipeline end to end.?

[RAG-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Agentic RAG?

[RAG-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is GraphRAG?

[RAG-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Retrieval-Augmented Generation?

[RAG-005] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a document loader?

[RAG-006] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is context injection?

[RAG-007] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is conversational RAG?

[RAG-008] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is document ingestion?

[RAG-009] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is metadata enrichment?

[RAG-010] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is multi-document RAG?

[RAG-011] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is multi-hop RAG?

[RAG-012] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is overlap and why is it used?

[RAG-013] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is query embedding?

[RAG-014] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is recursive character splitting?

[RAG-015] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is text splitting?

[RAG-016] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How is retrieval different from generation?

[RAG-017] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When should you combine RAG and fine-tuning?

[RAG-018] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When should you use GraphRAG?

[RAG-019] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why do we need RAG if an LLM already has knowledge?

TOPIC 45: DOCUMENT PARSING, CHUNKING AND INDEXING

Total: **9 Questions** | High Priority: **5** | Medium: **4** | Lower: **0**

[RAG-020] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are the indexing and query-time stages of RAG?

[RAG-021] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is chunking?

[RAG-022] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is fixed-size chunking?

[RAG-023] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is parent-child chunking?

[RAG-024] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is semantic chunking?

[RAG-025] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you choose chunk size?

[RAG-026] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What happens if chunks are too small?

[RAG-027] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What metadata should be stored with chunks?

[RAG-028] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why is chunking important?

TOPIC 46: EMBEDDING MODELS AND VECTOR DATABASES

Total: **28 Questions** | High Priority: **19** | Medium: **8** | Lower: **1**

[RAG-029] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How are text embeddings generated?

[RAG-030] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How do you update embeddings when documents change?

[RAG-031] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is BM25?

[RAG-032] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is HNSW?

[RAG-033] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is IVF?

[RAG-034] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is MRR?

[RAG-035] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is PQ/product quantization?

[RAG-036] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Precision@K?

[RAG-037] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Recall@K?

[RAG-038] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a vector database?

[RAG-039] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an embedding model?

[RAG-040] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an embedding?

[RAG-041] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is cosine similarity?

[RAG-042] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is dense retrieval?

[RAG-043] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is hybrid retrieval?

[RAG-044] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is nDCG?

[RAG-045] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is nearest-neighbor search?

[RAG-046] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is semantic similarity?

[RAG-047] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is sparse retrieval?

[RAG-048] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you evaluate retrieval quality?

[RAG-049] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle duplicate documents?

[RAG-050] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle multi-tenant vector stores?

[RAG-051] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you select an embedding model?

[RAG-052] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does embedding drift affect a production RAG system?

[RAG-053] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What does embedding dimensionality mean?

[RAG-054] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What happens if documents and queries use incompatible embedding models?

[RAG-055] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When does keyword search outperform vector search?

[RAG-056] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

Why use a vector database instead of a relational database? What are metadata filters?

TOPIC 47: SPARSE RETRIEVAL, BM25 AND HYBRID SEARCH

Total: **2 Questions** | High Priority: **1** | Medium: **1** | Lower: **0**

[RAG-057] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is hybrid search?

[RAG-058] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you combine BM25 and vector search?

TOPIC 48: RERANKING AND ADVANCED RETRIEVAL

Total: **2 Questions** | High Priority: **1** | Medium: **1** | Lower: **0**

[RAG-059] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is reranking?

[RAG-060] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why is reranking useful?

TOPIC 49: ADVANCED RAG, GRAPHRAG AND MULTI-HOP RAG

Total: **31 Questions** | High Priority: **16** | Medium: **15** | Lower: **0**

[RAG-061] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is HYDE?

[RAG-062] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Self-RAG?

[RAG-063] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is adaptive RAG?

[RAG-064] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is contextual compression?

[RAG-065] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is contextual retrieval?

[RAG-066] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is corrective RAG?

[RAG-067] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is filtered vector search?

[RAG-068] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is hierarchical retrieval?

[RAG-069] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is late interaction retrieval?

[RAG-070] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is metadata-based retrieval?

[RAG-071] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is multi-query retrieval?

[RAG-072] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is parent-document retrieval?

[RAG-073] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is query expansion?

[RAG-074] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is query rewriting?

[RAG-075] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is reciprocal rank fusion?

[RAG-076] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is routing in RAG?

[RAG-077] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you combine SQL retrieval and vector retrieval?

[RAG-078] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you debug high retrieval recall but poor answers?

[RAG-079] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you debug low retrieval recall?

[RAG-080] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle PDFs with images?

[RAG-081] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle code repositories with RAG?

[RAG-082] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle conflicting documents?

[RAG-083] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle scanned documents?

[RAG-084] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle stale documents?

[RAG-085] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle structured databases in RAG?

[RAG-086] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle tables in RAG?

[RAG-087] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you implement access-control-aware retrieval?

[RAG-088] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you prevent irrelevant chunks from entering the context?

[RAG-089] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you reduce RAG cost?

[RAG-090] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you reduce RAG latency?

[RAG-091] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you route queries across multiple indexes?

TOPIC 50: RAG EVALUATION AND TROUBLESHOOTING

Total: **31 Questions** | High Priority: **15** | Medium: **15** | Lower: **1**

[RAG-092] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is RAG evaluation?

[RAG-093] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is RAGAS?

[RAG-094] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a golden dataset?

[RAG-095] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an evaluation dataset?

[RAG-096] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is answer relevance?

[RAG-097] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is citation correctness?

[RAG-098] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is context precision?

[RAG-099] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is context recall?

[RAG-100] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is context relevance?

[RAG-101] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is faithfulness?

[RAG-102] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is generation evaluation?

[RAG-103] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is groundedness?

[RAG-104] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is pairwise evaluation?

[RAG-105] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is regression testing for LLM applications?

[RAG-106] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is retrieval evaluation?

[RAG-107] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do latency, cost, and quality trade off?

[RAG-108] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you create a golden set for RAG?

[RAG-109] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you evaluate a RAG system without human labels?

[RAG-110] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you evaluate retrieval independently from generation?

[RAG-111] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you debug a RAG system that hallucinates?

[RAG-112] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you detect retrieval drift?

[RAG-113] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you monitor RAG quality in production?

[RAG-114] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What if answers are incomplete?

[RAG-115] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What if answers are too verbose?

[RAG-116] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What if answers cite the wrong source?

[RAG-117] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What if retrieval is correct but the model ignores the context?

[RAG-118] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What if retrieval returns the wrong documents?

[RAG-119] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What if the model answers questions that are not supported by the documents?

[RAG-120] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When should LLM-as-a-judge be used?

[RAG-121] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Which RAG metrics would you track?

[RAG-122] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What are the limitations of LLM-as-a-judge?

TOPIC 51: HALLUCINATION REDUCTION, GROUNDING AND CITATIONS

Total: **2 Questions** | High Priority: **2** | Medium: **0** | Lower: **0**

[RAG-123] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Can RAG completely eliminate hallucination?

[RAG-124] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How does RAG reduce hallucination?

PART VIII
AGENTIC AI AND ORCHESTRATION

TOPIC 52: AI AGENT FUNDAMENTALS

Total: **30 Questions** | High Priority: **21** | Medium: **9** | Lower: **0**

[AGT-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are the core components of an agent?

[AGT-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is ReAct?

[AGT-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a bounded agent?

[AGT-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is agent memory?

[AGT-005] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an AI agent?

[AGT-006] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an agent loop?

[AGT-007] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an autonomous agent?

[AGT-008] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is deterministic orchestration?

[AGT-009] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is episodic memory?

[AGT-010] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is function calling?

[AGT-011] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is human-in-the-loop?

[AGT-012] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is long-term memory?

[AGT-013] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is planning?

[AGT-014] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is procedural memory?

[AGT-015] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is reflection?

[AGT-016] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is self-correction?

[AGT-017] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is semantic memory?

[AGT-018] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is short-term memory?

[AGT-019] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is state in an agent?

[AGT-020] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is tool calling?

[AGT-021] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is working memory?

[AGT-022] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you limit tool calls?

[AGT-023] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you set agent budgets?

[AGT-024] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you stop infinite agent loops?

[AGT-025] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How is an agent different from a chatbot?

[AGT-026] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How is an agent different from a normal LLM chain?

[AGT-027] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What causes agent loops?

[AGT-028] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When should an agent ask for human approval?

[AGT-029] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When should you prefer a workflow over an agent?

[AGT-030] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why can agents become expensive?

TOPIC 53: TOOL CALLING AND STRUCTURED OUTPUTS

Total: **19 Questions** | High Priority: **3** | Medium: **16** | Lower: **0**

[AGT-031] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a tool allowlist?

[AGT-032] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is least privilege for agents?

[AGT-033] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is runtime validation?

[AGT-034] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you audit tool calls?

[AGT-035] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you design tools for an agent?

[AGT-036] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle destructive tools?

[AGT-037] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle duplicate tool execution?

[AGT-038] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle tool timeouts?

[AGT-039] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle tool versioning?

[AGT-040] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you make tool calls idempotent?

[AGT-041] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you prevent an agent from calling unauthorized tools?

[AGT-042] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you require approval before sensitive actions?

[AGT-043] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you validate tool arguments?

[AGT-044] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does LLM tool calling work?

[AGT-045] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How should tool errors be returned to the model?

[AGT-046] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What happens when a tool call fails?

[AGT-047] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What information does a tool schema contain?

[AGT-048] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why should tools have narrow responsibilities?

[AGT-049] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why use Pydantic models for tool arguments?

TOPIC 54: MCP AND TOOL INTEGRATION

Total: **8 Questions** | High Priority: **4** | Medium: **4** | Lower: **0**

[MCP-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is Model Context Protocol (MCP)?

[MCP-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an MCP prompt?

[MCP-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an MCP resource?

[MCP-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an MCP tool?

[MCP-005] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you secure an MCP server?

[MCP-006] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does MCP differ from ordinary function calling?

[MCP-007] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What problem does MCP solve?

[MCP-008] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When would you use MCP?

TOPIC 55: LANGCHAIN FRAMEWORK

Total: **20 Questions** | High Priority: **16** | Medium: **4** | Lower: **0**

[LGC-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are AIMessage, HumanMessage, and SystemMessage?

[LGC-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are prompts in LangChain?

[LGC-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is LCEL?

[LGC-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is LangChain?

[LGC-005] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is LlamaIndex?

[LGC-006] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a Runnable?

[LGC-007] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a chain?

[LGC-008] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a retriever?

[LGC-009] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a tool in LangChain?

[LGC-010] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an edge?

[LGC-011] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is batching?

[LGC-012] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is checkpointing?

[LGC-013] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is conditional routing?

[LGC-014] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is graph state?

[LGC-015] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is middleware?

[LGC-016] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is tool calling in LangChain?

[LGC-017] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you persist agent state?

[LGC-018] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you debug an agent graph?

[LGC-019] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What problem does LangChain solve?

[LGC-020] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What problem does LlamaIndex solve?

TOPIC 56: LANGGRAPH, STATE, NODES, EDGES AND CHECKPOINTING

Total: **6 Questions** | High Priority: **3** | Medium: **3** | Lower: **0**

[LGC-021] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a node in LangGraph?

[LGC-022] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an interrupt in LangGraph?

[LGC-023] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is streaming? What is LangGraph?

[LGC-024] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How does LangGraph support human-in-the-loop?

[LGC-025] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When would you use LangGraph for production?

[LGC-026] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why use LangGraph instead of a simple chain?

TOPIC 57: MULTI-AGENT SYSTEMS AND ORCHESTRATION

Total: **16 Questions** | High Priority: **7** | Medium: **8** | Lower: **1**

[AGT-050] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a critic agent?

[AGT-051] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a judge agent?

[AGT-052] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a multi-agent system?

[AGT-053] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a planner-worker architecture?

[AGT-054] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a router agent?

[AGT-055] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a supervisor agent?

[AGT-056] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is agent delegation?

[AGT-057] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do agents communicate?

[AGT-058] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you control multi-agent cost?

[AGT-059] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you evaluate individual agents?

[AGT-060] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you evaluate the whole multi-agent workflow?

[AGT-061] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you prevent multi-agent loops?

[AGT-062] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you secure multi-agent systems?

[AGT-063] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When is a single agent better than multiple agents?

[AGT-064] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When should you use multiple agents?

[AGT-065] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What are the disadvantages of multi-agent systems?

TOPIC 58: AGENT MEMORY AND CONTEXT MANAGEMENT

Total: **14 Questions** | High Priority: **3** | Medium: **11** | Lower: **0**

[AGT-066] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is context compaction?

[AGT-067] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is memory in an AI agent?

[AGT-068] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is memory retrieval?

[AGT-069] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you decide what information enters the next model call?

[AGT-070] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you delete or correct memory?

[AGT-071] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle stale memories?

[AGT-072] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you isolate memory between users?

[AGT-073] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you manage context-window limits in agents?

[AGT-074] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you prevent memory poisoning?

[AGT-075] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you rank memories?

[AGT-076] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you store agent state?

[AGT-077] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you summarize long conversations?

[AGT-078] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What should an agent remember?

[AGT-079] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why should memory be separated from the prompt?

TOPIC 59: AI SAFETY, SECURITY AND GUARDRAILS

Total: **24 Questions** | High Priority: **9** | Medium: **15** | Lower: **0**

[AGT-080] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How do you detect prompt injection inside retrieved content?

[AGT-081] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What are guardrails?

[AGT-082] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is PII?

[AGT-083] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is data exfiltration?

[AGT-084] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is defense in depth for AI systems?

[AGT-085] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is input validation?

[AGT-086] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is jailbreak?

[AGT-087] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is least privilege in agent design?

[AGT-088] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is output validation?

[AGT-089] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you audit AI actions?

[AGT-090] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you design safe fallback behavior?

[AGT-091] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you handle malicious documents in RAG?

[AGT-092] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you handle unsafe model outputs?

[AGT-093] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you implement approval gates?

[AGT-094] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you isolate tenant data?

[AGT-095] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you prevent PII leakage?

[AGT-096] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you prevent SQL injection through an LLM-generated query?

[AGT-097] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you prevent an agent from sending unauthorized emails?

[AGT-098] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you protect against tool poisoning?

[AGT-099] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you protect secrets from an LLM?

[AGT-100] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you safely execute generated code?

[AGT-101] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
How do you sandbox tool execution?

[AGT-102] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
Should API keys ever be placed in prompts?

[AGT-103] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>
Why are guardrails needed for agents?

PART IX
LLMOPS AND AI SYSTEM DESIGN

TOPIC 60: LLMOPS AND OBSERVABILITY

Total: **20 Questions** | High Priority: **9** | Medium: **10** | Lower: **1**

[LLMO-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

How do you monitor hallucinations?

[LLMO-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is LLMOps?

[LLMO-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is an LLM span?

[LLMO-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is caching?

[LLMO-005] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is continuous batching?

[LLMO-006] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is distributed tracing?

[LLMO-007] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is observability for agents?

[LLMO-008] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is semantic caching?

[LLMO-009] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is tracing?

[LLMO-010] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you deploy open-weight models?

[LLMO-011] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you design for high availability?

[LLMO-012] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you detect model regressions?

[LLMO-013] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle rate limits?

[LLMO-014] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you implement retries?

[LLMO-015] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you monitor tool failures?

[LLMO-016] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you version prompts, models, datasets, and indexes?

[LLMO-017] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What metrics should you monitor?

[LLMO-018] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

What should you log in an LLM application?

[LLMO-019] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why can retries be dangerous for tools?

[LLMO-020] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

What is exponential backoff?

TOPIC 61: EVALUATION FRAMEWORKS AND TESTING

Total: **1 Questions** | High Priority: **0** | Medium: **1** | Lower: **0**

[LLMO-021] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you benchmark an LLM service?

TOPIC 62: LATENCY, TOKEN COSTS, CACHING AND OPTIMIZATION

Total: **3 Questions** | High Priority: **0** | Medium: **3** | Lower: **0**

[LLMO-022] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you monitor latency?

[LLMO-023] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you monitor token usage?

[LLMO-024] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you reduce inference cost?

TOPIC 63: RELIABILITY, RETRIES AND PRODUCTION MONITORING

Total: **4 Questions** | High Priority: **1** | Medium: **3** | Lower: **0**

[LLMO-025] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is circuit breaking?

[LLMO-026] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you handle model provider outages?

[LLMO-027] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How do you implement fallbacks across models?

[LLMO-028] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

When should you retry an LLM call?

TOPIC 64: MODEL ROUTING AND INFERENCE ARCHITECTURE

Total: **3 Questions** | High Priority: **2** | Medium: **1** | Lower: **0**

[LLMO-029] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a model gateway?

[LLMO-030] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is vLLM?

[LLMO-031] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Why use model routing?

TOPIC 65: END-TO-END RAG SYSTEM DESIGN

Total: **8 Questions** | High Priority: **0** | Medium: **8** | Lower: **0**

[SYS-001] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a RAG system for frequently changing documents.

[SYS-002] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a RAG system with strict data isolation.

[SYS-003] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a legal-document RAG system with citations.

[SYS-004] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a multi-tenant RAG platform.

[SYS-005] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a multimodal RAG system.

[SYS-006] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a production GraphRAG system.

[SYS-007] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design an enterprise RAG system end to end.

[SYS-008] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you scale a RAG system from 1K to 10M users?

TOPIC 66: AGENTIC WORKFLOW SYSTEM DESIGN

Total: **10 Questions** | High Priority: **0** | Medium: **10** | Lower: **0**

[SYS-009] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a SQL analytics agent.

[SYS-010] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a browser-use research agent.

[SYS-011] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a customer-support AI agent.

[SYS-012] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a customer-support multi-agent system.

[SYS-013] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a financial research agent.

[SYS-014] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a multi-agent customer-support platform.

[SYS-015] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a research multi-agent system.

[SYS-016] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design an AI coding agent.

[SYS-017] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design an email automation agent with human approval.

[SYS-018] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you design asynchronous agent execution?

TOPIC 67: ENTERPRISE AI ARCHITECTURE AND SCALABILITY

Total: **9 Questions** | High Priority: **0** | Medium: **9** | Lower: **0**

[SYS-019] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a document question-answering system for 10 million documents.

[SYS-020] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a model-routing service.

[SYS-021] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design a prompt-management platform.

[SYS-022] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design an AI system with 99.9% availability.

[SYS-023] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design an LLM evaluation platform.

[SYS-024] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Design an LLM gateway supporting multiple providers.

[SYS-025] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you control token cost at scale?

[SYS-026] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you design a human-in-the-loop approval system?

[SYS-027] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you design rate limiting for an LLM API?

PART X
PRACTICAL INTERVIEW ROUNDS

TOPIC 68: CODING AND IMPLEMENTATION QUESTIONS

Total: **20 Questions** | High Priority: **1** | Medium: **18** | Lower: **1**

[CODE-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Implement top-k retrieval from a list of embeddings.

[CODE-002] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Create a Pydantic model for tool arguments.

[CODE-003] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Create an evaluation loop for a RAG golden dataset.

[CODE-004] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you mock an LLM during testing?

[CODE-005] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you test an agent deterministically?

[CODE-006] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

How would you unit-test an LLM application?

[CODE-007] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Implement a simple ReAct-style agent loop.

[CODE-008] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Implement a timeout around a tool call.

[CODE-009] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Implement a tool registry in Python.

[CODE-010] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Implement async parallel tool calls.

[CODE-011] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Implement cosine similarity between two vectors.

[CODE-012] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Implement reciprocal rank fusion.

[CODE-013] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Validate an LLM-generated JSON response with Pydantic.

[CODE-014] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Write a Python function to chunk a document with overlap.

[CODE-015] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Write a function to truncate context while preserving high-priority chunks.

[CODE-016] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Write a simple BM25-style keyword retrieval function.

[CODE-017] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Write a simple conversation-memory manager.

[CODE-018] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Write a token-budget limiter for an agent.

[CODE-019] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Write code to remove duplicate documents.

[CODE-020] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

Write a retry wrapper with exponential backoff.

TOPIC 69: SCENARIO-BASED DATA SCIENCE QUESTIONS

Total: **25 Questions** | High Priority: **2** | Medium: **11** | Lower: **12**

[CASE-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Fine-tuning improves tone but increases hallucination. What happened?

[CASE-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Your vector database becomes slow at scale. What do you investigate?

[CASE-003] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

An agent creates an infinite loop. How do you detect and stop it?

[CASE-004] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Your LLM outputs invalid JSON 5% of the time. How do you improve reliability?

[CASE-005] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Your LLM provider has an outage. What is your fallback?

[CASE-006] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Your RAG has good retrieval metrics but users dislike answers. Why?

[CASE-007] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Your agent repeatedly calls the same tool. How do you fix it?

[CASE-008] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Your application suddenly doubles token usage. How do you investigate?

[CASE-009] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Your documents are updated hourly. How do you keep RAG fresh?

[CASE-010] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Your embedding model changed. What migration strategy would you use?

[CASE-011] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Your evaluation score improves but production quality falls. Why?

[CASE-012] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Your fine-tuned model is worse than the base model. How do you debug it?

[CASE-013] ★★☆ PRIORITY: MEDIUM | VIEW ANSWER >>

Your retriever has poor recall. How do you improve it?

[CASE-014] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

An attacker puts malicious instructions into a PDF used by RAG. What happens and how do you defend?

[CASE-015] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

Two tenants can see each other's retrieved documents. What do you do immediately?

[CASE-016] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

Users report that answers cite irrelevant sources. How do you fix citation quality?

[CASE-017] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

Your RAG retrieves relevant documents but answers are wrong. What do you investigate?

[CASE-018] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

Your RAG works in development but fails under production traffic. What do you investigate?

[CASE-019] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

Your agent has access to 30 tools and chooses incorrectly. How do you improve tool selection?

[CASE-020] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

Your agent latency is 10 seconds and the target is 2 seconds. What do you optimize?

[CASE-021] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

Your agent makes an irreversible action without approval. What changes do you make?

[CASE-022] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

Your context is too large and the model misses important facts. How do you fix it?

[CASE-023] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

Your customer asks for an answer not present in the knowledge base. What should the system do?

[CASE-024] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

Your model is cheap but inaccurate; another is accurate but expensive. How do you route requests?

[CASE-025] ★☆☆ PRIORITY: LOWER | VIEW ANSWER >>

Your multi-agent system costs 5x more than a single agent. How do you decide whether it is justified?

TOPIC 70: RAPID-FIRE COMPARISONS

Total: **37 Questions** | High Priority: **37** | Medium: **0** | Lower: **0**

[COMP-001] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Accuracy vs latency?

[COMP-002] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Agent vs workflow: what is the trade-off?

[COMP-003] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

BM25 vs vector search?

[COMP-004] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Bi-encoder vs cross-encoder?

[COMP-005] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Chunking vs semantic chunking?

[COMP-006] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Conversation history vs memory?

[COMP-007] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Cosine similarity vs Euclidean distance?

[COMP-008] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

DPO vs RLHF?

[COMP-009] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Dense vs sparse retrieval?

[COMP-010] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Fine-tuning vs RAG?

[COMP-011] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Fine-tuning vs continued pretraining?

[COMP-012] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Fine-tuning vs prompt engineering?

[COMP-013] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Full fine-tuning vs LoRA?

[COMP-014] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Human-in-the-loop vs fully autonomous agents?

[COMP-015] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

LangChain agent vs LangGraph?

[COMP-016] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

LangChain vs LlamaIndex?

[COMP-017] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

LangGraph vs LangChain agents?

[COMP-018] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

LoRA vs QLoRA?

[COMP-019] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

MCP server vs MCP client?

[COMP-020] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

MCP vs REST APP

[COMP-021] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Offline eval vs online eval?

[COMP-022] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Prompt engineering vs context engineering?

[COMP-023] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Quality vs cost?

[COMP-024] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

RAG vs fine-tuning?

[COMP-025] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

RAG vs long-context prompting?

[COMP-026] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

RLHF vs DPO?

[COMP-027] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Redis vs SQL vs vector DB for memory?

[COMP-028] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

SFT vs DPO?

[COMP-029] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Shared state vs message passing?

[COMP-030] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Short-term vs long-term memory?

[COMP-031] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Single agent vs multi-agent?

[COMP-032] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Streaming vs batching?

[COMP-033] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Tool calling vs MCP?

[COMP-034] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

TypedDict vs Pydantic for tool inputs?

[COMP-035] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Vector DB vs SQL database?

[COMP-036] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

What is a base model vs an instruction-tuned model?

[COMP-037] ★★★ PRIORITY: HIGH | VIEW ANSWER >>

Workflow vs agent?